

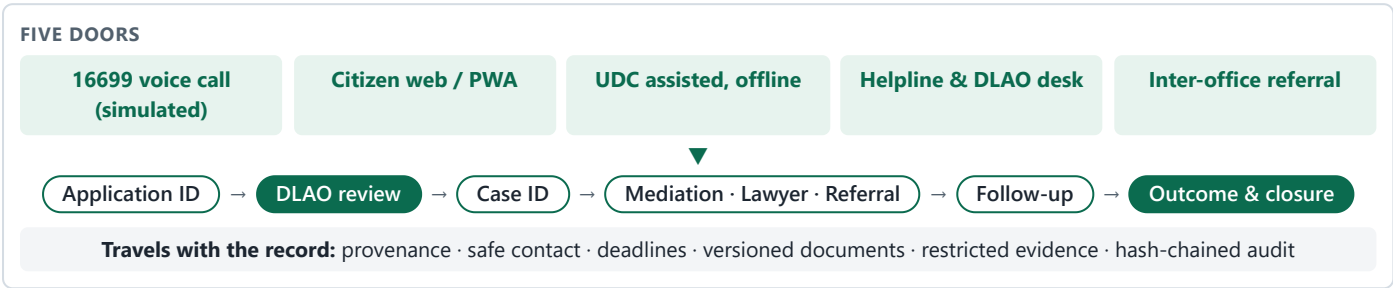
Five Doors, One Record: Integrated Digital Legal Aid System (DLAS)

Solution paper · ADLASB Grand Finale · 26 September 2026 · all data fictional

One system, not 23 islands: every citizen door, provider role and technical feature works on **one authoritative Application/Case record**. AI suggests; authorised humans decide.

1. ARCHITECTURE AND INTEGRATION MAP

React/Vite PWA → Express 5 API (MVC) → MongoDB with transactions. Submission issues an **Application ID**; only a DLAO officer's acceptance creates the **Case ID**. Facts, consent, safe contact, tasks, documents, referrals, mediation, lawyer work and audit attach to that record. The server enforces role, office and assignment. Optional Groq AI always has a manual fallback.



2. THE 23 PROBLEMS AND OUR SOLUTIONS

#	PROBLEM	SOLUTION	STATUS
PART A: CITIZENS			
A1	Moyuri: unsafe contact, no NID, brother reports	Ripon's facts stay REPRESENTATIVE_REPORTED and identity incomplete until she confirms. Safe-contact profile enforced; if someone else answers, nothing is disclosed. Her correction is appended as APPLICANT_CONFIRMED; his version stays.	Partial Officer-attested correction; no self-service screen
A2	Ripon: blind representative	Voice or keypad call with 38 recorded Bangla prompts, read-back, ID and PIN spoken; no CAPTCHA, visual OTP or PDF. Representation stays pending.	Partial Blind-user test pending
A3	Nabila: urgent, sensitive images	Urgency reasons with evidence; officer sets priority. Restricted evidence needs a named grant or open referral; every access logged. Tracked referral; no acknowledgement → follow-up task.	✓
A4	Nuching: Marma, cannot read, weak network	Applicant, helper, translator and typist recorded separately; Marma original beside translation, read back; helper's phone never becomes applicant contact; drafts survive network loss.	✓
A5	Malek: long case, shop phone, silent lawyer	"Ask by Voice": he speaks his ID and PIN and hears stage, hearing date and next step, after a private-listening check. Failed contacts logged; overdue lawyer updates alert the DLAO.	✓
PART B: PROVIDERS			
B1	DLAO officer: scattered registers	One office queue: flag reasons, ageing, owner, next action, search; priority override needs a reason and is audited.	✓
B2	Mediator	Registration, scheduling (in-person, remote or hybrid), notices, documents, attendance, session and outcome, with T7 drafting and T11 signing.	✓
B3	16699 helpline agent	Exact-ID lookup; generic status only after one-time code, verification and safe channel; new intakes join the same record.	✓
B4	UDC entrepreneur	Free-service notice, checklist, consent, provenance, safe contact; access ends at DLAO review or 24 hours.	✓
B5	Panel lawyer	Worklist, accept/decline, hearings, update schedule, reports, fee claims, overdue reminders; client contact only after acceptance.	✓
B6	Receiving DLAO	Package: reason, history, documents, responsible officer, expected action, deadline; accept or return with reason; both offices see status.	✓
B7	Case-support staff	Structured record, search, versions, tasks, audit reconstruction, reports from data captured once; restricted evidence hidden.	✓

#	PROBLEM	SOLUTION	STATUS
PART C: TECHNICAL CHALLENGES			
T1	Lawyer change, inactivity, payment	Change request → DLAO review → replacement; stage payment history; two missed updates → review and temporary assignment hold, not a misconduct finding.	✓
T2	Jurisdiction ping-pong	Two returns escalate; transfers blocked until a DLAO records a reasoned routing decision.	✓
T3	One incident, many applicants	Three Cases linked, never merged; common document uploaded once, shared by reference.	✓
T4	Duplicate / fraud risk	Multi-attribute fuzzy matching over 12 profiles; side-by-side human review with reason; never merges or rejects; two traps stay separate.	✓
T5	Bangla intake agent	AI fills only approved fields, re-validated and marked AI-inferred; unclear safety answers go to an officer with the transcript.	✓
T6	Document agent	Six documents → briefing quoting source version and line; flags a missing and an unreadable item; officer approves.	✓
T7	Settlement drafting	Maintenance, property and labour templates; AI-filled fields and inconsistencies marked; human review before signing.	✓
T8	Multi-agent triage	Categorisation, process/safety and urgency components; disagreements shown; no names sent to AI; DLAO decides. Five cases, one conflict.	✓
T9	Offline sync, integrity	Encrypted drafts with temporary and mutation IDs sync once; conflicts go to a human; hash chain verified.	✓
T10	Low-bandwidth PWA	Installable; caches static shell only; light mode (automatic on Save-Data/2G) cuts call start 468 → 304 ms at 50 KB/s.	✓
T11	Offline e-signature	Parties sign separately with P-256 keys, one offline then synced; mediator last; a changed copy fails verification.	✓

✓ built, integrated and covered by automated tests.

3. HUMAN SAFEGUARDS

- **AI proposes, humans decide:** AI never sets eligibility, priority, route, lawyer, outcome or provenance.
- Every override and decision needs a reason and is audited.
- Scores and alerts open reviews; they never reject, merge or find misconduct.
- Signatures are not a decree; CLAO certification is a separate authorised step.
- Demo thresholds await law-team approval.

4. RESILIENCE AND FAILURE MODEL

Unknown person answers	No disclosure; safer follow-up
Referral unacknowledged	Follow-up task, audit event
Network drop	Encrypted draft; no duplicate on sync
Offline edit conflict	Human chooses, with reason
Lawyer misses two updates	Review and assignment hold
AI unavailable	Deterministic or keyboard path
Signed text altered	Verification fails
Wrong role opens evidence	Denied and logged

Audit is tamper-evident (SHA-256 chain), not tamper-proof.

5. FIVE SUCCESS INDICATORS

INDICATOR	MEASURE AND PROTOTYPE EVIDENCE	TARGET
Safe-contact compliance	Contact attempts with zero unauthorised disclosure; unknown-person test passes.	100%
One-record continuity	Journeys where all doors and roles share one Application/Case ID; six jury flows.	100%
Accessible completion	Ripon, Nuching and Malek tasks done without an excluded step; blind-user test pending.	≥ 90%
Tracked responsibility	Referrals, lawyer updates and mediation steps with owner, deadline and escalation.	100%
Resilient recovery	Failure tests recovered without loss or duplicates; 12 bad-day tests automated.	100%

Stated limits: 16699 telephony, SMS/USSD, government ID, payments and notifications are simulated or not connected. Pending: blind-user and real-caller validation, scanned-PDF OCR, multi-server timers, law-team approval. Targets are proposals.